

4(a)(i)	distance (in a specified direction of particle/point on wave) from the equilibrium position	B1
4(a)(ii)	the maximum distance (of particle/point on wave) from the equilibrium position or the maximum displacement (of particle/point on wave)	B1
4(b)	$I \propto A^2$	C1
	$I_R / I = (3.6 - 1.2)^2 / (1.2)^2$ resultant intensity = $4.0I$	A1
4(c)(i)	as wave(s) pass through the slit(s)	B1
	wave(s) spread (into geometric shadow)	B1
4(c)(ii)	$n\lambda = d \sin \theta$	C1
	$3\lambda = d \sin 90^\circ$ or $3\lambda = d$	C1
	$d = 3 \times 630 \times 10^{-9}$ $= 1.9 \times 10^{-6} \text{ m}$	A1
4(c)(iii)	wavelength of blue light is shorter (than 540 nm/630 nm/wavelengths of original light)	M1
	(so) third order diffraction maximum is produced	A1